

11.5. Linearity, monotonicity, and additivity

The integral has the following three basic properties.

(1) Linearity:

$$\int_a^b cf = c \int_a^b f, \quad \int_a^b (f + g) = \int_a^b f + \int_a^b g.$$

(2) Monotonicity: if $f \leq g$, then

$$\int_a^b f \leq \int_a^b g.$$

(3) Additivity: if $a < c < b$, then

$$\int_a^c f + \int_c^b f = \int_a^b f.$$

These properties are analogous to the corresponding properties of sums (or convergent series):

$$\begin{aligned} \sum_{k=1}^n ca_k &= c \sum_{k=1}^n a_k, & \sum_{k=1}^n (a_k + b_k) &= \sum_{k=1}^n a_k + \sum_{k=1}^n b_k; \\ \sum_{k=1}^n a_k &\leq \sum_{k=1}^n b_k & \text{if } a_k &\leq b_k; \\ \sum_{k=1}^m a_k + \sum_{k=m+1}^n a_k &= \sum_{k=1}^n a_k. \end{aligned}$$

In this section, we prove these properties and derive a few of their consequences.

11.5.1. Linearity. We begin by proving the linearity. First we prove linearity with respect to scalar multiplication and then linearity with respect to sums.

Theorem 11.32. If $f : [a, b] \rightarrow \mathbb{R}$ is integrable and $c \in \mathbb{R}$, then cf is integrable and

$$\int_a^b cf = c \int_a^b f.$$

Proof. Suppose that $c \geq 0$. Then for any set $A \subset [a, b]$, we have

$$\sup_A cf = c \sup_A f, \quad \inf_A cf = c \inf_A f,$$

so $U(cf; P) = cU(f; P)$ for every partition P . Taking the infimum over the set Π of all partitions of $[a, b]$, we get

$$U(cf) = \inf_{P \in \Pi} U(cf; P) = \inf_{P \in \Pi} cU(f; P) = c \inf_{P \in \Pi} U(f; P) = cU(f).$$

Similarly, $L(cf; P) = cL(f; P)$ and $L(cf) = cL(f)$. If f is integrable, then

$$U(cf) = cU(f) = cL(f) = L(cf),$$

which shows that cf is integrable and

$$\int_a^b cf = c \int_a^b f.$$

Now consider $-f$. Since

$$\sup_A(-f) = -\inf_A f, \quad \inf_A(-f) = -\sup_A f,$$

we have

$$U(-f; P) = -L(f; P), \quad L(-f; P) = -U(f; P).$$

Therefore

$$\begin{aligned} U(-f) &= \inf_{P \in \Pi} U(-f; P) = \inf_{P \in \Pi} [-L(f; P)] = -\sup_{P \in \Pi} L(f; P) = -L(f), \\ L(-f) &= \sup_{P \in \Pi} L(-f; P) = \sup_{P \in \Pi} [-U(f; P)] = -\inf_{P \in \Pi} U(f; P) = -U(f). \end{aligned}$$

Hence, $-f$ is integrable if f is integrable and

$$\int_a^b (-f) = -\int_a^b f.$$

Finally, if $c < 0$, then $c = -|c|$, and a successive application of the previous results shows that cf is integrable with $\int_a^b cf = c \int_a^b f$. $\square$

Next, we prove the linearity of the integral with respect to sums. If f, g are bounded, then $f + g$ is bounded and

$$\sup_I(f + g) \leq \sup_I f + \sup_I g, \quad \inf_I(f + g) \geq \inf_I f + \inf_I g.$$

It follows that

$$\operatorname{osc}_I(f + g) \leq \operatorname{osc}_I f + \operatorname{osc}_I g,$$

so $f + g$ is integrable if f, g are integrable. In general, however, the upper (or lower) sum of $f + g$ needn't be the sum of the corresponding upper (or lower) sums of f and g . As a result, we don't get

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g$$

simply by adding upper and lower sums. Instead, we prove this equality by estimating the upper and lower integrals of $f + g$ from above and below by those of f and g .

Theorem 11.33. If $f, g : [a, b] \rightarrow \mathbb{R}$ are integrable functions, then $f + g$ is integrable, and

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g.$$

Proof. We first prove that if $f, g : [a, b] \rightarrow \mathbb{R}$ are bounded, but not necessarily integrable, then

$$U(f + g) \leq U(f) + U(g), \quad L(f + g) \geq L(f) + L(g).$$

Suppose that $P = \{I_1, I_2, \dots, I_n\}$ is a partition of $[a, b]$. Then

$$\begin{aligned} U(f+g; P) &= \sum_{k=1}^n \sup_{I_k} (f+g) \cdot |I_k| \\ &\leq \sum_{k=1}^n \sup_{I_k} f \cdot |I_k| + \sum_{k=1}^n \sup_{I_k} g \cdot |I_k| \\ &\leq U(f; P) + U(g; P). \end{aligned}$$

Let $\epsilon > 0$. Since the upper integral is the infimum of the upper sums, there are partitions Q, R such that

$$U(f; Q) < U(f) + \frac{\epsilon}{2}, \quad U(g; R) < U(g) + \frac{\epsilon}{2},$$

and if P is a common refinement of Q and R , then

$$U(f; P) < U(f) + \frac{\epsilon}{2}, \quad U(g; P) < U(g) + \frac{\epsilon}{2}.$$

It follows that

$$U(f+g) \leq U(f+g; P) \leq U(f; P) + U(g; P) < U(f) + U(g) + \epsilon.$$

Since this inequality holds for arbitrary $\epsilon > 0$, we must have $U(f+g) \leq U(f) + U(g)$.

Similarly, we have $L(f+g; P) \geq L(f; P) + L(g; P)$ for all partitions P , and for every $\epsilon > 0$, we get $L(f+g) > L(f) + L(g) - \epsilon$, so $L(f+g) \geq L(f) + L(g)$.

For integrable functions f and g , it follows that

$$U(f+g) \leq U(f) + U(g) = L(f) + L(g) \leq L(f+g).$$

Since $U(f+g) \geq L(f+g)$, we have $U(f+g) = L(f+g)$ and $f+g$ is integrable. Moreover, there is equality throughout the previous inequality, which proves the result. $\square$

Although the integral is linear, the upper and lower integrals of non-integrable functions are not, in general, linear.

Example 11.34. Define $f, g : [0, 1] \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x \in [0, 1] \cap \mathbb{Q}, \\ 0 & \text{if } x \in [0, 1] \setminus \mathbb{Q}, \end{cases} \quad g(x) = \begin{cases} 0 & \text{if } x \in [0, 1] \cap \mathbb{Q}, \\ 1 & \text{if } x \in [0, 1] \setminus \mathbb{Q}. \end{cases}$$

That is, f is the Dirichlet function and $g = 1 - f$. Then

$$U(f) = U(g) = 1, \quad L(f) = L(g) = 0, \quad U(f+g) = L(f+g) = 1,$$

so

$$U(f+g) < U(f) + U(g), \quad L(f+g) > L(f) + L(g).$$

The product of integrable functions is also integrable, as is the quotient provided it remains bounded. Unlike the integral of the sum, however, there is no way to express the integral of the product $\int fg$ in terms of $\int f$ and $\int g$.

Theorem 11.35. If $f, g : [a, b] \rightarrow \mathbb{R}$ are integrable, then $fg : [a, b] \rightarrow \mathbb{R}$ is integrable. If, in addition, $g \neq 0$ and $1/g$ is bounded, then $f/g : [a, b] \rightarrow \mathbb{R}$ is integrable.

Proof. First, we show that the square of an integrable function is integrable. If f is integrable, then f is bounded, with $|f| \leq M$ for some $M \geq 0$. For all $x, y \in [a, b]$, we have

$$|f^2(x) - f^2(y)| = |f(x) + f(y)| \cdot |f(x) - f(y)| \leq 2M|f(x) - f(y)|.$$

Taking the supremum of this inequality over $x, y \in I \subset [a, b]$ and using Proposition 11.8, we get that

$$\sup_I(f^2) - \inf_I(f^2) \leq 2M \left[\sup_I f - \inf_I f \right],$$

meaning that

$$\operatorname{osc}_I(f^2) \leq 2M \operatorname{osc}_I f.$$

It follows from Proposition 11.25 that f^2 is integrable if f is integrable.

Since the integral is linear, we then see from the identity

$$fg = \frac{1}{4} [(f+g)^2 - (f-g)^2]$$

that fg is integrable if f, g are integrable. We remark that the trick of representing a product as a difference of squares isn't a new one: the ancient Babylonian apparently used this identity, together with a table of squares, to compute products.

In a similar way, if $g \neq 0$ and $|1/g| \leq M$, then

$$\left| \frac{1}{g(x)} - \frac{1}{g(y)} \right| = \frac{|g(y) - g(x)|}{|g(x)g(y)|} \leq M^2 |g(x) - g(y)|.$$

Taking the supremum of this equation over $x, y \in I \subset [a, b]$, we get

$$\sup_I \left(\frac{1}{g} \right) - \inf_I \left(\frac{1}{g} \right) \leq M^2 \left[\sup_I g - \inf_I g \right],$$

meaning that $\operatorname{osc}_I(1/g) \leq M^2 \operatorname{osc}_I g$, and Proposition 11.25 implies that $1/g$ is integrable if g is integrable. Therefore $f/g = f \cdot (1/g)$ is integrable. $\square$

11.5.2. Monotonicity. Next, we prove the monotonicity of the integral.

Theorem 11.36. Suppose that $f, g : [a, b] \rightarrow \mathbb{R}$ are integrable and $f \leq g$. Then

$$\int_a^b f \leq \int_a^b g.$$

Proof. First suppose that $f \geq 0$ is integrable. Let P be the partition consisting of the single interval $[a, b]$. Then

$$L(f; P) = \inf_{[a, b]} f \cdot (b - a) \geq 0,$$

so

$$\int_a^b f \geq L(f; P) \geq 0.$$

If $f \geq g$, then $h = f - g \geq 0$, and the linearity of the integral implies that

$$\int_a^b f - \int_a^b g = \int_a^b h \geq 0,$$

which proves the theorem. $\square$

One immediate consequence of this theorem is the following simple, but useful, estimate for integrals.

Theorem 11.37. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is integrable and

$$M = \sup_{[a,b]} f, \quad m = \inf_{[a,b]} f.$$

Then

$$m(b-a) \leq \int_a^b f \leq M(b-a).$$

Proof. Since $m \leq f \leq M$ on $[a, b]$, Theorem 11.36 implies that

$$\int_a^b m \leq \int_a^b f \leq \int_a^b M,$$

which gives the result. $\square$

This estimate also follows from the definition of the integral in terms of upper and lower sums, but once we've established the monotonicity of the integral, we don't need to go back to the definition.

A further consequence is the intermediate value theorem for integrals, which states that a continuous function on a compact interval is equal to its average value at some point in the interval.

Theorem 11.38. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then there exists $x \in [a, b]$ such that

$$f(x) = \frac{1}{b-a} \int_a^b f.$$

Proof. Since f is a continuous function on a compact interval, the extreme value theorem (Theorem 7.37) implies it attains its maximum value M and its minimum value m . From Theorem 11.37,

$$m \leq \frac{1}{b-a} \int_a^b f \leq M.$$

By the intermediate value theorem (Theorem 7.44), f takes on every value between m and M , and the result follows. $\square$

As shown in the proof of Theorem 11.36, given linearity, monotonicity is equivalent to positivity,

$$\int_a^b f \geq 0 \quad \text{if } f \geq 0.$$

We remark that even though the upper and lower integrals aren't linear, they are monotone.

Proposition 11.39. If $f, g : [a, b] \rightarrow \mathbb{R}$ are bounded functions and $f \leq g$, then

$$U(f) \leq U(g), \quad L(f) \leq L(g).$$

Proof. From Proposition 11.1, we have for every interval $I \subset [a, b]$ that

$$\sup_I f \leq \sup_I g, \quad \inf_I f \leq \inf_I g.$$

It follows that for every partition P of $[a, b]$, we have

$$U(f; P) \leq U(g; P), \quad L(f; P) \leq L(g; P).$$

Taking the infimum of the upper inequality and the supremum of the lower inequality over P , we get that $U(f) \leq U(g)$ and $L(f) \leq L(g)$. $\square$

We can estimate the absolute value of an integral by taking the absolute value under the integral sign. This is analogous to the corresponding property of sums:

$$\left| \sum_{k=1}^n a_k \right| \leq \sum_{k=1}^n |a_k|.$$

Theorem 11.40. If f is integrable, then $|f|$ is integrable and

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

Proof. First, suppose that $|f|$ is integrable. Since

$$-|f| \leq f \leq |f|,$$

we get from Theorem 11.36 that

$$-\int_a^b |f| \leq \int_a^b f \leq \int_a^b |f|, \quad \text{or} \quad \left| \int_a^b f \right| \leq \int_a^b |f|.$$

To complete the proof, we need to show that $|f|$ is integrable if f is integrable. For $x, y \in [a, b]$, the reverse triangle inequality gives

$$||f(x)| - |f(y)|| \leq |f(x) - f(y)|.$$

Using Proposition 11.8, we get that

$$\sup_I |f| - \inf_I |f| \leq \sup_I f - \inf_I f,$$

meaning that $\text{osc}_I |f| \leq \text{osc}_I f$. Proposition 11.25 then implies that $|f|$ is integrable if f is integrable. $\square$

In particular, we immediately get the following basic estimate for an integral.

Corollary 11.41. If $f : [a, b] \rightarrow \mathbb{R}$ is integrable and $M = \sup_{[a, b]} |f|$, then

$$\left| \int_a^b f \right| \leq M(b - a).$$

Finally, we prove a useful positivity result for the integral of continuous functions.

Proposition 11.42. If $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function such that $f \geq 0$ and $\int_a^b f = 0$, then $f = 0$.

Proof. Suppose for contradiction that $f(c) > 0$ for some $a \leq c \leq b$. For definiteness, assume that $a < c < b$. (The proof is similar if c is an endpoint.) Then, since f is continuous, there exists $\delta > 0$ such that

$$|f(x) - f(c)| \leq \frac{f(c)}{2} \quad \text{for } c - \delta \leq x \leq c + \delta,$$

where we choose δ small enough that $c - \delta > a$ and $c + \delta < b$. It follows that

$$f(x) = f(c) + f(x) - f(c) \geq f(c) - |f(x) - f(c)| \geq \frac{f(c)}{2}$$

for $c - \delta \leq x \leq c + \delta$. Using this inequality and the assumption that $f \geq 0$, we get

$$\int_a^b f = \int_a^{c-\delta} f + \int_{c-\delta}^{c+\delta} f + \int_{c+\delta}^b f \geq 0 + \frac{f(c)}{2} \cdot 2\delta + 0 > 0.$$

This contradiction proves the result. $\square$

The assumption that $f \geq 0$ is, of course, required, otherwise the integral of the function may be zero due to cancellation.

Example 11.43. The function $f : [-1, 1] \rightarrow \mathbb{R}$ defined by $f(x) = x$ is continuous and nonzero, but $\int_{-1}^1 f = 0$.

Continuity is also required; for example, the discontinuous function in Example 11.14 is nonzero, but its integral is zero.

11.5.3. Additivity. Finally, we prove additivity. This property refers to additivity with respect to the interval of integration, rather than linearity with respect to the function being integrated.

Theorem 11.44. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ and $a < c < b$. Then f is Riemann integrable on $[a, b]$ if and only if it is Riemann integrable on $[a, c]$ and $[c, b]$. Moreover, in that case,

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Proof. Suppose that f is integrable on $[a, b]$. Then, given $\epsilon > 0$, there is a partition P of $[a, b]$ such that $U(f; P) - L(f; P) < \epsilon$. Let $P' = P \cup \{c\}$ be the refinement of P obtained by adding c to the endpoints of P . (If $c \in P$, then $P' = P$.) Then $P' = Q \cup R$ where $Q = P' \cap [a, c]$ and $R = P' \cap [c, b]$ are partitions of $[a, c]$ and $[c, b]$ respectively. Moreover,

$$U(f; P') = U(f; Q) + U(f; R), \quad L(f; P') = L(f; Q) + L(f; R).$$

It follows that

$$\begin{aligned} U(f; Q) - L(f; Q) &= U(f; P') - L(f; P') - [U(f; R) - L(f; R)] \\ &\leq U(f; P) - L(f; P) < \epsilon, \end{aligned}$$

which proves that f is integrable on $[a, c]$. Exchanging Q and R , we get the proof for $[c, b]$.

Conversely, if f is integrable on $[a, c]$ and $[c, b]$, then there are partitions Q of $[a, c]$ and R of $[c, b]$ such that

$$U(f; Q) - L(f; Q) < \frac{\epsilon}{2}, \quad U(f; R) - L(f; R) < \frac{\epsilon}{2}.$$

Let $P = Q \cup R$. Then

$$U(f; P) - L(f; P) = U(f; Q) - L(f; Q) + U(f; R) - L(f; R) < \epsilon,$$

which proves that f is integrable on $[a, b]$.

Finally, if f is integrable, then with the partitions P, Q, R as above, we have

$$\begin{aligned} \int_a^b f &\leq U(f; P) = U(f; Q) + U(f; R) \\ &< L(f; Q) + L(f; R) + \epsilon \\ &< \int_a^c f + \int_c^b f + \epsilon. \end{aligned}$$

Similarly,

$$\begin{aligned} \int_a^b f &\geq L(f; P) = L(f; Q) + L(f; R) \\ &> U(f; Q) + U(f; R) - \epsilon \\ &> \int_a^c f + \int_c^b f - \epsilon. \end{aligned}$$

Since $\epsilon > 0$ is arbitrary, we see that $\int_a^b f = \int_a^c f + \int_c^b f$. □

We can extend the additivity property of the integral by defining an oriented Riemann integral.

Definition 11.45. If $f : [a, b] \rightarrow \mathbb{R}$ is integrable, where $a < b$, and $a \leq c \leq b$, then

$$\int_b^a f = - \int_a^b f, \quad \int_c^c f = 0.$$

With this definition, the additivity property in Theorem 11.44 holds for all $a, b, c \in \mathbb{R}$ for which the oriented integrals exist. Moreover, if $|f| \leq M$, then the estimate in Corollary 11.41 becomes

$$\left| \int_a^b f \right| \leq M|b - a|$$

for all $a, b \in \mathbb{R}$ (even if $a \geq b$).

The oriented Riemann integral is a special case of the integral of a differential form. It assigns a value to the integral of a one-form $f dx$ on an oriented interval.